

Strings, Character Arrays and std::string in C++

> **Definition to Remember**

1. Concept Overview & Fundamentals

Strings, Character Arrays and std::string in C++ provides modern high-performance capabilities combining low-level hardware control with high-level abstractions, object-oriented paradigms, and Standard Template Library (STL) utilities.

In modern C++ (C++11 through C++20):

2. Technical Architecture & Memory Mechanics

C++ Runtime Execution Architecture

```text

### Modern C++ Feature Comparison

| Modern C++ Paradigm | Legacy Equivalent | Engineering Advantage | Performance Impact |

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3. Core Syntax & Implementation Patterns

Modern C++ Template Structure

```
```cpp
// Fast I/O and standard namespace configuration

std::cout << "Executing C++ Module: Strings" << "\n";

return 0;
}
```
```

4. Real-World Applications & System Engineering

- **Quantitative Finance & HFT:** High-frequency trading platforms use modern C++ for microsecond-level execution.

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5. Comprehensive Solved Problems & Code Walkthroughs

Problem 1: break keyword loop

```
Problem Description & Objective:

C++ Code Implementation:
```cpp
// Implementation details...
```

/*

- ===

#include <iostream>

#include

// Use standard library namespace globally

using

// --- Main Execution Entry Point ---

int main

// Loop execution block

while (

// Evaluate conditional decision logic

if (input

// Output formatted data to console stream

cout <

// Return zero code indicating successful program finish

return

Problem 2: date format

****Problem Description & Objective:****

****C++ Code Implementation:****

/*

#include <iostream>

// Use standard library namespace globally

// --- Main Execution Entry Point ---

// Output formatted data to console stream

// Return zero code indicating successful program finish

return

Problem 3: fgets puts****Problem Description & Objective:****

Implem

****C++ Code Implementation:****

```cpp

/\*

- ===

#include &lt;iostream&gt;

#includ

// Use standard library namespace globally

using

// --- Main Execution Entry Point ---

int mai

// Output formatted data to console stream

cout &lt;

// Output formatted data to console stream

cout &lt;

```
// Return zero code indicating successful program finish
```

```
return
```

```

```

#### Problem 4: login system

**\*\*Problem Description & Objective:\*\***

Implemen

**\*\*C++ Code Implementation:\*\***

```
```cpp
```

```
/*
```

```
- ===
```

```
#include <iostream>
```

```
#includ
```

```
// Use standard library namespace globally
```

```
using n
```

```
// --- Main Execution Entry Point ---
```

```
int mai
```

```
string username, password;
```

```
// Output formatted data to console stream
```

```
cout <
```

```
// Output formatted data to console stream
```

```
cout <<
```

```
// Evaluate conditional decision logic
```

```
if (user
```

```
// Return zero code indicating successful program finish
```

```
return
```

```
---
```

Problem 5: password checker

```
**Problem Description & Objective:**
```

```
Implem
```

```
**C++ Code Implementation:**
```

```
```cpp
```

```
/*
```

```
- ===
```

```
#include <iostream>
```

```
// Use standard library namespace globally
```

```
// --- Main Execution Entry Point ---
```

```
do {
```

```
// Evaluate conditional decision logic
```

```
// Output formatted data to console stream
```

```
// Return zero code indicating successful program finish
```

```

```

### **Problem 6: reverse string function**

**\*\*Problem Description & Objective:\*\***

**\*\*C++ Code Implementation:\*\***

```
/*
```

```
#include <iostream>
```

```
// Use standard library namespace globally
```

```
void reverseStringInPlace(string &str) {
```

```
// --- Main Execution Entry Point ---
```

```
reverseStringInPlace(text);
```

```
// Output formatted data to console stream
```

```
// Return zero code indicating successful program finish
```

```

```

**Problem 7: string palindrome****\*\*Problem Description & Objective:\*\***

Implemen

**\*\*C++ Code Implementation:\*\***

```cpp

/*

- ===

#include <iostream>

#includ

// Use standard library namespace globally

using n

bool isPalindromeString(const string &str) {

int left

// Loop execution block

while (

```
// --- Main Execution Entry Point ---
```

```
int main
```

```
// Evaluate conditional decision logic
```

```
if (isPa
```

```
// Return zero code indicating successful program finish
```

```
return
```

```
---
```

Problem 8: tic tac toe board

****Problem Description & Objective:****

Implemen

****C++ Code Implementation:****

```
```cpp
```

```
/*
```

```
- ===
```

```
#include <iostream>
```

```
// Use standard library namespace globally
```

```
// Declare dynamic C++ std::vector container
```

```
// --- Main Execution Entry Point ---
```

```
// Output formatted data to console stream
```

```
board[0][0] = 'X';
```

```
// Output formatted data to console stream
```

```
// Return zero code indicating successful program finish
```

---

### Problem 9: trim string

**\*\*Problem Description & Objective:\*\***

**\*\*C++ Code Implementation:\*\***

/\*

#include <iostream>

// Use standard library namespace globally

string trimString(const string &str) {

// --- Main Execution Entry Point ---

```
// Output formatted data to console stream
```

```
// Return zero code indicating successful program finish
```

```

```

### Problem 10: uppercase string

**\*\*Problem Description & Objective:\*\***

**\*\*C++ Code Implementation:\*\***

```
/*
```

```
#include <iostream>
```

```
// Use standard library namespace globally
```

```
// --- Main Execution Entry Point ---
```

```
// Loop execution block
```

```
// Output formatted data to console stream
```

```
// Return zero code indicating successful program finish
```

```

```

## 6. Exam Scoring Points & Memory Revision

```
> Must-Write Points
```

```
> Quick Recall
```